ABDUL MANAN
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ROS

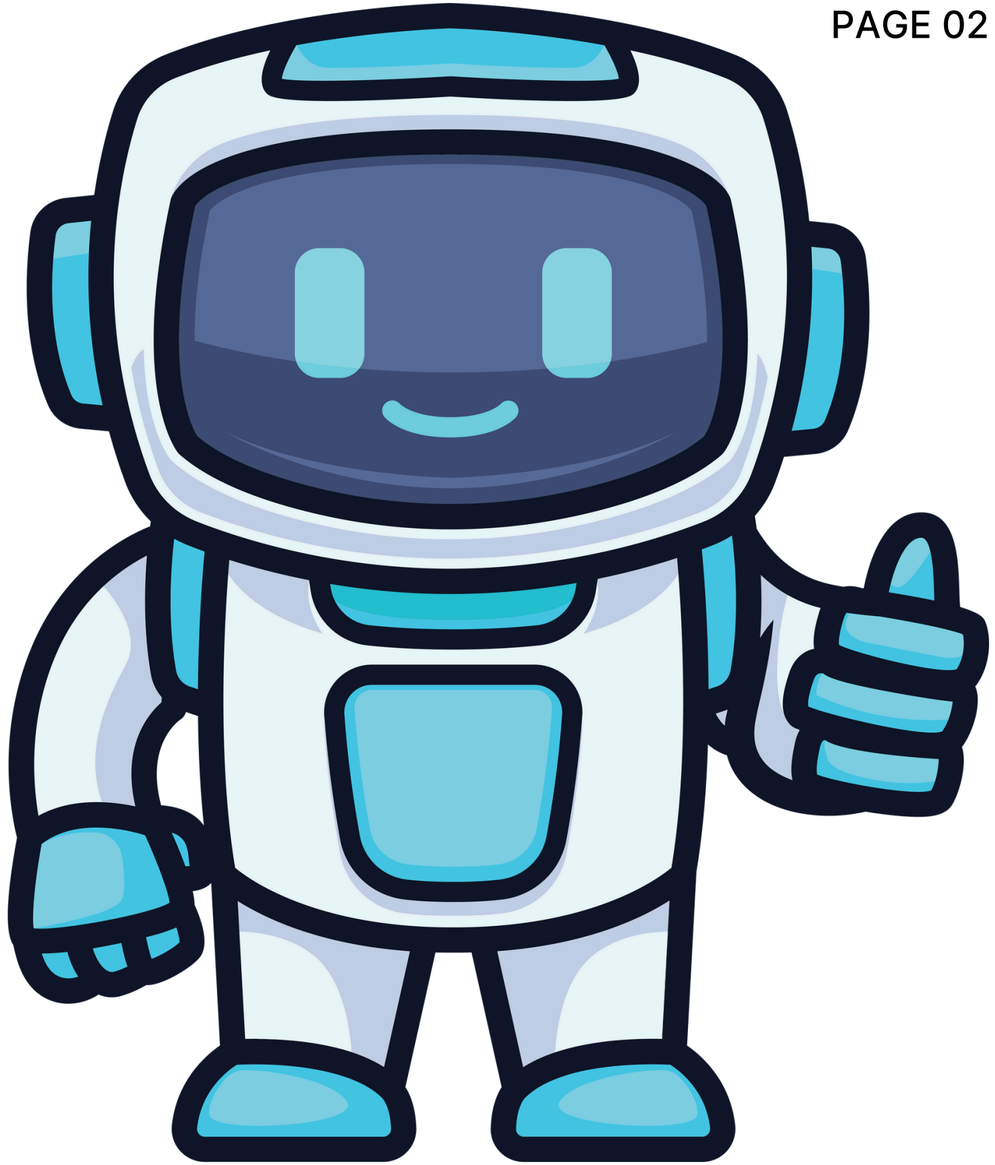
WHERE ROBOTS LEARN TO WORK TOGETHER

Architecture, Applications & Microcontroller Compatibility

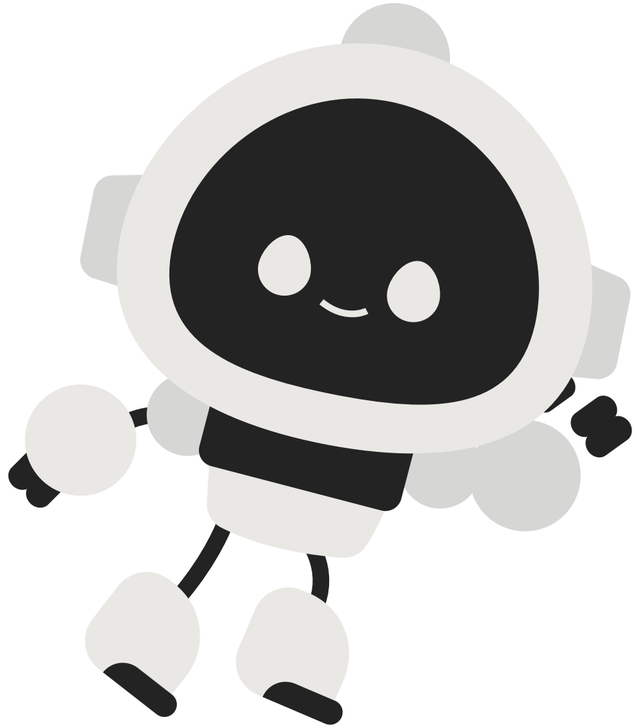


WHAT IS ROS?

- ROS is an open-source middleware used to build and control robotic systems.
- It provides tools, libraries, and communication frameworks for robotic applications.
- Not an operating system — it runs on top of Linux (mostly Ubuntu)
- Enables modular development through nodes, topics, services, and packages.
- Helps integrate sensors, actuators, AI algorithms, and controllers easily.
- Widely used in research, industry, and autonomous systems.



INTRODUCTION & MOTIVATION



Robotics systems have become increasingly complex, requiring advanced software frameworks. To tackle these challenges, ROS came, initially developed by two PhD students from Stanford University.

01

ROS 1

Became widely used but lacked determinism, security & industrial readiness.

02

ROS 2

Addresses these limitations using DDS, modular architecture & RTOS compatibility.

ROS-I VS ROS-II

	ROS-I	ROS-II
Architecture	single point of failure	robust, peer-to-peer communication
Security	No native security	authentication, encryption & access control
Embedded Systems	Not suitable for microcontrollers	Supports microcontrollers via micro-ROS
Multi-Platform & Multi-Robot	Constrained; weak multi-robot support	Multi-robot, multi-platform (Linux, Windows, macOS,)

ROS-II ARCHITECTURE OVERVIEW









01

DDS MIDDLEWARE:

Real-time, QoS profiles, peer-to-peer, no master node

02

NODES, TOPICS, SERVICES, ACTIONS:

Nodes, Topics, Services, Actions: Communication patterns

03

LIFECYCLE NODES:

Managed states for safety & predictability

04

COLCON BUILD SYSTEM:

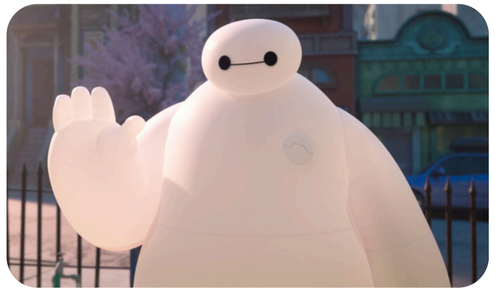
Modern multi-package build tool



AUTONOMOUS VEHICLES



INDUSTRIAL AUTOMATION



HEALTHCARE ROBOTICS



DRONES & UAVS

ROS2 APPLICATIONS

01

AUTONOMOUS VEHICLES

Sensorfusion, Navigation, Autoware

02

DRONES & UAVS

Distributed control, microROSintegration

03

INDUSTRIAL AUTOMATION

AMRs, cobots, manipulators

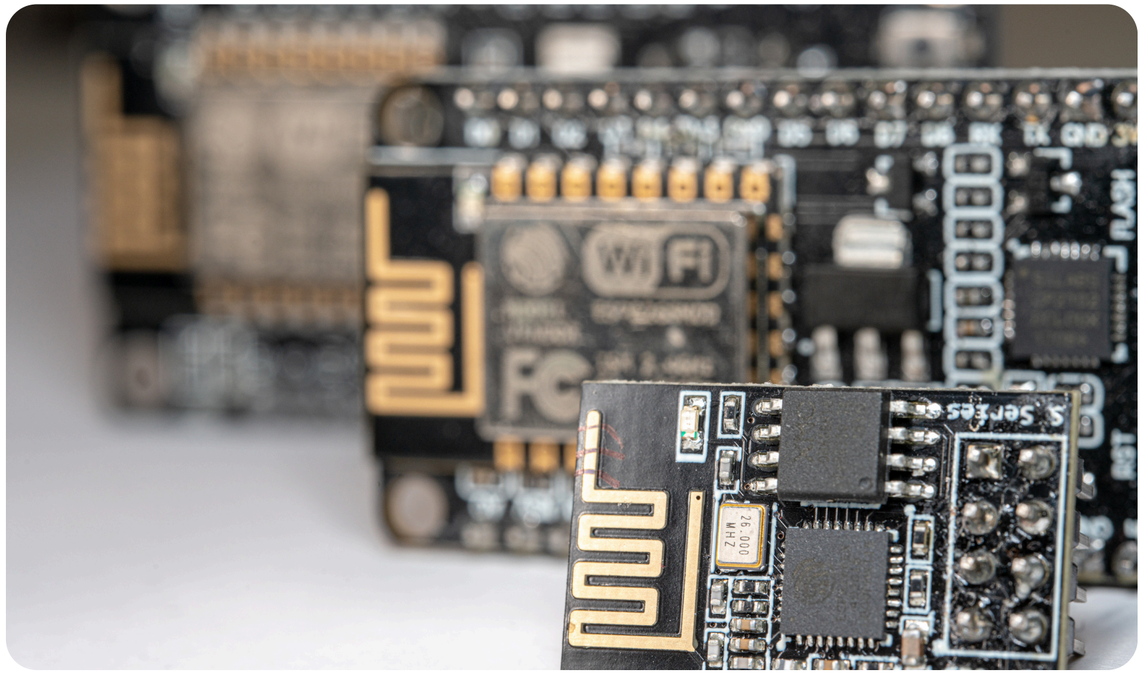
04

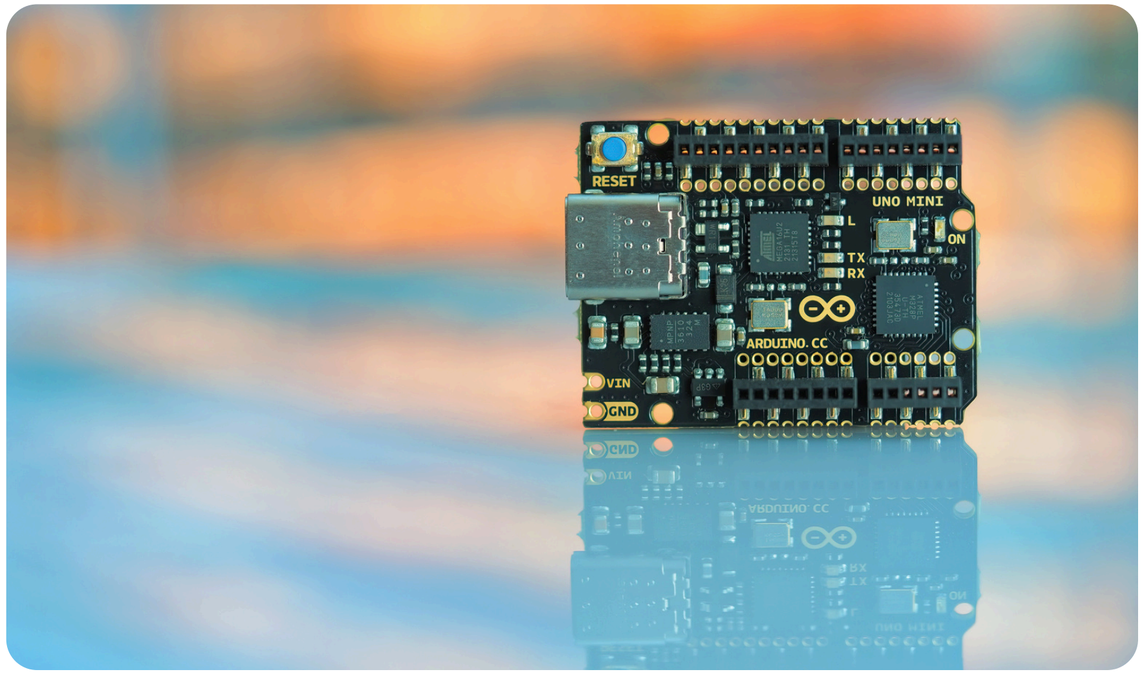
HEALTHCARE ROBOTICS

Assistive robots, surgical systems

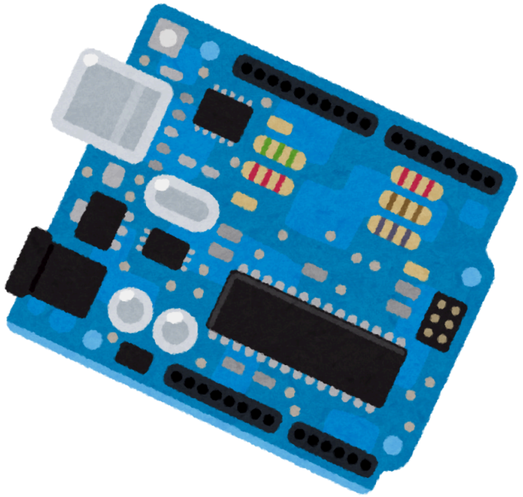
COMPATIBILITY WITH MICROCONTROLLERS (MICRO-ROS)

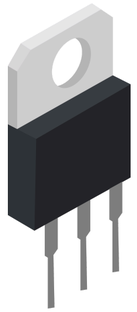
- Full ROS 2 cannot run on MCUs → microROSsolves this.
- Runs on FreeRTOS, Zephyr, NuttX,Arduino framework.
- Architecture: MCU (client) ↔ micro-ROS Agent↔ROS2network

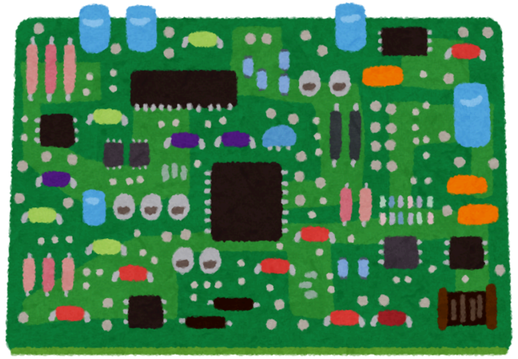


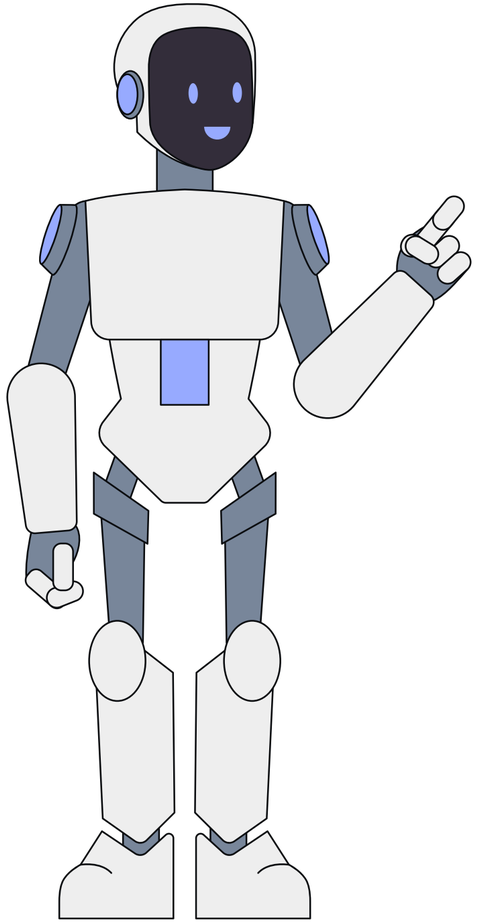


SUPPORTED MCUS: ESP32 STM32 / NUCLEO RP2040 (RASPBERRY PI PICO) ARDUINO PORTENTA H7









CONCLUSION

- ROS 2 is a major upgrade over ROS 1:realtime, secure, scalable.
- Widely adopted across research & industry.
- Micro-ROS enables embedded/IoT devices to join ROS 2 systems.
- Future directions: micro-ROS expansion, AI integration, improved real-time execution.

ROS: WHERE ROBOTS LEARN TO WORK TOGETHER

THANK YOU FOR YOUR ATTENTION

